

# GreenSure

## Software Requirements Specification

Carbon Footprint Tracking & Green Insurance Platform

### 1. Introduction

#### 1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the GreenSure platform. It is intended to serve as a comprehensive reference for development, testing, and future enhancements of the system.

#### 1.2 Project Overview

GreenSure is a full-stack web application that calculates a user's annual carbon footprint and links that score directly to an insurance premium discount. The core motivation is to address the global carbon emissions problem by giving individuals and businesses a direct financial incentive to live and operate more sustainably.

"The greener you live, the less you pay."

GreenSure is not just an insurance application. It is a behaviour change platform that uses money as the motivator to push users toward a greener lifestyle year on year.

#### 1.3 Scope

The GreenSure platform covers the following scope:

- Annual carbon declaration submission by households and MSMEs
- Field agent physical verification of declared data with GPS proof
- Automated carbon score calculation using the CarbonScoreService engine
- Zone-based insurance premium discount allocation
- Year-on-year improvement bonus for users who reduce their carbon score
- Role-based access control across four roles: User, Field Agent, Admin, and System
- Automated email notifications for all key events
- Three Spring Schedulers running automated background processes

## 1.4 Definitions & Acronyms

| Term / Acronym  | Definition                                                                            |
|-----------------|---------------------------------------------------------------------------------------|
| SRS             | Software Requirements Specification                                                   |
| SPA             | Single Page Application — Angular frontend loaded once, routes handled client-side    |
| RBAC            | Role-Based Access Control — access enforced based on user role                        |
| JWT             | JSON Web Token — stateless authentication token                                       |
| MSME            | Micro, Small and Medium Enterprise — a business user type in GreenSure                |
| HH              | Household — a residential user type in GreenSure                                      |
| CO <sub>2</sub> | Carbon Dioxide — unit of carbon emission measurement                                  |
| CarbonScore     | The verified annual CO <sub>2</sub> score assigned to a user after agent verification |
| Zone            | Green Champion, Improver, or Defaulter — classification based on carbon score         |
| Scheduler       | Spring @Scheduled background task running on fixed intervals                          |
| SMTP            | Simple Mail Transfer Protocol — used for all outgoing emails                          |

## 2. Overall Description

### 2.1 The Problem Being Solved

Global carbon emissions are rising every year. A significant contribution comes from individual and business lifestyle choices — energy consumption, vehicles driven, and operational waste. Despite widespread awareness, there is no direct financial motivation for individuals to change their behaviour.

Insurance companies have historically priced premiums based solely on postcode and age, completely ignoring a customer's environmental behaviour. A household with rooftop solar and an EV pays the exact same premium as a household running diesel generators and three petrol cars. There is no reward for making greener choices and no transparency on how premiums are calculated.

#### Key Market Gap:

73% of insurance customers want personalised premiums.

Less than 8% of insurers use behavioural data in pricing.

Companies globally are committing to net zero emissions targets —  
yet individual insurance products do nothing to accelerate that goal.

### 2.2 Product Perspective

GreenSure fills this gap by building a platform where a user's verified carbon footprint is the primary driver of their insurance pricing. The platform creates a habit loop — every year a user sees their score, sees their discount, and sees exactly what they need to change to save more money next year. This loop is what drives real, measurable lifestyle change.

### 2.3 User Classes and Characteristics

#### 2.3.1 User (Household / MSME)

Regular citizens or small business owners who submit their annual carbon declaration, receive a verified carbon score, and get insurance premium discounts based on that score. Users can be of two subtypes:

- Household — individual residential users tracking personal carbon footprint
- MSME — small business owners tracking business operational emissions

#### 2.3.2 Field Agent

On-ground employees who physically visit homes and businesses to verify the carbon data submitted by users. Agents are the credibility layer of GreenSure — agent-verified data always overrides user-declared values.

#### 2.3.3 Admin

Platform administrators who manage users, agents, declarations, assignments, and overall platform health. Admins have full visibility and control over all platform data.

### 2.3.4 System (Schedulers)

Three automated Spring Schedulers that run continuously in the background without any human trigger:

- AssignmentEngine — runs every 30 minutes, auto-assigns agents to unassigned declarations
- DeadlineMonitor — runs every hour, detects missed deadlines and reassigns automatically
- RenewalEngine — runs on the 1st of every month at 9AM, sends renewal reminder emails

## 2.4 Operating Environment

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|                    |                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Backend:</b>    | Spring Boot 4.0, Java 21, Spring Security, JWT, Spring Data JPA, Hibernate, H2 (dev) / PostgreSQL (production), JavaMailSender / SMTP |
| <b>Frontend:</b>   | Angular 21, TypeScript, Tailwind CSS, Lucide Icons, Angular Signals, RxJS, Auth Guards, HTTP Interceptors                             |
| <b>Auth:</b>       | Stateless JWT — no sessions on server. Token carries userId, email, role, userType. Validated by JwtAuthFilter on every request.      |
| <b>Deployment:</b> | Backend: AWS / Railway. Frontend: Vercel / Netlify.                                                                                   |

## 3. System Architecture

### 3.1 Layered Architecture

GreenSure follows a 4-layer N-Tier architecture with strict separation of concerns between each layer:

| Layer                 | Technology                                        | Responsibility                                                              |
|-----------------------|---------------------------------------------------|-----------------------------------------------------------------------------|
| Layer 1 — Frontend    | Angular 21, TypeScript, Tailwind CSS              | SPA, role-based views, Auth Guards, HTTP Interceptors, user interaction     |
| Layer 2 — Controllers | Spring Boot, Spring Security, @PreAuthorize       | REST endpoints, JWT validation, RBAC enforcement, request/response handling |
| Layer 3 — Services    | Spring @Service, Spring Scheduler, JavaMailSender | Business logic, carbon score engine, schedulers, email dispatch             |
| Layer 4 — Data        | Spring Data JPA, Hibernate, H2 / PostgreSQL       | Entities, repositories, persistence, ORM mappings                           |

### 3.2 Security Architecture

GreenSure implements stateless JWT-based authentication with backend RBAC enforcement:

- Every request passes through JwtAuthFilter before reaching any controller
- JWT token carries: userId, email, role (USER / AGENT / ADMIN), userType (HOUSEHOLD / MSME)
- @PreAuthorize annotations enforce role-based access at the method level
- Frontend Auth Guards and HTTP Interceptors handle UX-level RBAC and auto-attach tokens
- Frontend RBAC controls what the user sees. Backend RBAC controls what they can access. Both are independent.

### 3.3 Frontend Structure

The Angular frontend is organized under src/app/ as follows:

```
src/app/  
  features/  
    auth/      login, register, forgot-password, reset-password  
    user/      dashboard, profile, policies, my-policies, score-history,  
               recommendations, notifications, welcome  
    declaration/ declaration-start, declaration-fill, declaration-vehicles,  
               declaration-review, declaration-history  
    agent/     dashboard, task-detail, verify, performance, notifications  
    admin/     dashboard, users, user-detail, agents, create-agent,  
               agent-detail, declarations, assignments
```

shared/

components/ navbar, sidebar, alerts, badges, forms, inputs, charts

core/

guards/ AuthGuard, RoleGuard

interceptors/ JwtInterceptor, ErrorInterceptor

services/ AuthService and all feature services

## 4. Functional Requirements

### 4.1 Authentication Module

| Req ID     | Description                                                                                                                                                      | Priority | Role   |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-AUTH-01 | User registers with full name, email, password, user type (HOUSEHOLD or MSME), and profile details. System creates user account and sends welcome email.         | High     | All    |
| FR-AUTH-02 | User logs in with email and password. System validates credentials, generates JWT token, and returns token with user details.                                    | High     | All    |
| FR-AUTH-03 | JWT token is validated on every request by JwtAuthFilter before any controller is reached.                                                                       | High     | System |
| FR-AUTH-04 | User submits their email to the forgot password form. System generates a secure UUID token, saves it with 15-minute expiry, and sends a reset link to the email. | High     | User   |
| FR-AUTH-05 | User clicks reset link, lands on reset password page with token in URL. System validates token and expiry. User sets a new password. System clears the token.    | High     | User   |
| FR-AUTH-06 | Login and registration forms show red border on invalid fields and auto-scroll to the first error on submit.                                                     | Medium   | User   |

### 4.2 Carbon Declaration Module

| Req ID    | Description                                                                                                                                                                                                       | Priority | Role   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-DEC-01 | User submits an annual carbon declaration through a multi-step form capturing energy usage (kWh), transport km, vehicle details, and lifestyle / operations data. Household and MSME forms have different fields. | High     | User   |
| FR-DEC-02 | System enforces one declaration per user per year. Duplicate submissions are rejected with a custom exception.                                                                                                    | High     | System |
| FR-DEC-03 | Declaration status follows this state machine: DRAFT → SUBMITTED → UNDER_VERIFICATION → VERIFIED. If rejected by agent: REJECTED.                                                                                 | High     | System |
| FR-DEC-04 | If declaration is in DRAFT status, user can edit and save it. If SUBMITTED, UNDER_VERIFICATION, or VERIFIED, the form is read-only.                                                                               | High     | User   |
| FR-DEC-05 | If declaration is REJECTED, user can edit and resubmit. Maximum 3 resubmissions allowed. System tracks resubmission_count. On exceeding limit, throws MaxResubmissionLimitExceededException.                      | High     | User   |
| FR-DEC-06 | Declaration history page shows all past declarations. Clicking View Details shows details inline or in a modal — no navigation to a separate page.                                                                | Medium   | User   |
| FR-DEC-07 | User can add, edit, and remove vehicles from their declaration. Vehicles are stored as a list linked to the declaration.                                                                                          | High     | User   |

| GreenSure Platform |                                                                        | Software Requirements Specification |        |
|--------------------|------------------------------------------------------------------------|-------------------------------------|--------|
| Req ID             | Description                                                            | Priority                            | Role   |
| FR-DEC-08          | On successful submission, system sends email confirmation to the user. | Medium                              | System |

### 4.3 Agent Assignment Module

| Req ID    | Description                                                                                                                                                                                                                                                                      | Priority | Role   |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-ASN-01 | AssignmentEngine scheduler runs every 30 minutes. It finds all SUBMITTED declarations with no active assignment, matches an available agent by PIN code proximity, creates an AgentAssignment record, sets deadline to now + 72 hours, and updates status to UNDER_VERIFICATION. | High     | System |
| FR-ASN-02 | Admin can manually assign any available agent to any SUBMITTED declaration. System creates assignment, sets assignedBy to ADMIN, and sends email to agent.                                                                                                                       | High     | Admin  |
| FR-ASN-03 | Admin can reassign a declaration from one agent to another. Old assignment is marked REASSIGNED. New assignment is created. Both agents receive email notifications.                                                                                                             | High     | Admin  |
| FR-ASN-04 | Admin can cancel an active assignment. Assignment is marked CANCELLED. Declaration status reverts to SUBMITTED. Agent receives cancellation email.                                                                                                                               | High     | Admin  |
| FR-ASN-05 | DeadlineMonitor scheduler runs every hour. If an agent misses the 72-hour deadline, it auto-reassigns the declaration and applies a strike to the agent.                                                                                                                         | High     | System |
| FR-ASN-06 | Three strikes on an agent account can result in suspension.                                                                                                                                                                                                                      | Medium   | Admin  |
| FR-ASN-07 | assignedBy field stores SYSTEM for scheduler assignments and ADMIN for manual assignments.                                                                                                                                                                                       | Medium   | System |

### 4.4 Agent Verification Module

| Req ID    | Description                                                                                                                      | Priority | Role   |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-VER-01 | Agent views their assigned verification tasks. Each task shows declaration details, deadline, and user information.              | High     | Agent  |
| FR-VER-02 | Agent starts an assignment which logs GPS coordinates at the time of starting.                                                   | High     | Agent  |
| FR-VER-03 | Agent can CONFIRM declared values, MODIFY incorrect values on-site, or REJECT the declaration with a mandatory rejection reason. | High     | Agent  |
| FR-VER-04 | Agent-corrected values always override user-declared values in the carbon score calculation.                                     | High     | System |
| FR-VER-05 | On rejection, agent must provide a text rejection reason. System updates status to REJECTED and notifies the user.               | High     | Agent  |
| FR-VER-06 | Agent receives 24-hour deadline reminder email before their assignment deadline.                                                 | Medium   | System |



| GreenSure Platform |                                                                                      | Software Requirements Specification |       |
|--------------------|--------------------------------------------------------------------------------------|-------------------------------------|-------|
| Req ID             | Description                                                                          | Priority                            | Role  |
| FR-VER-07          | Agent can upload and attach documents or photos as part of the verification process. | Medium                              | Agent |

## 4.5 Carbon Score Calculation Engine

The CarbonScoreService calculates the carbon score in 8 sequential steps immediately after agent verification is submitted:

| Step | Name                            | Description                                                                                                                         |
|------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Duplicate Guard                 | Check no existing score for this declaration. Throw DuplicateScoreException if found.                                               |
| 2    | Load Verified Data              | Load declaration data applying agent corrections over user-declared values.                                                         |
| 3    | Energy CO <sub>2</sub>          | Electricity CO <sub>2</sub> = units (kWh) x 0.82 x 12 (annualised).                                                                 |
| 4    | Transport CO <sub>2</sub>       | Sum of vehicle km driven + public transport usage converted to CO <sub>2</sub> equivalent.                                          |
| 5    | Lifestyle / Ops CO <sub>2</sub> | Additional emissions from household lifestyle or MSME operations data.                                                              |
| 6    | Per Capita                      | Total CO <sub>2</sub> divided by number of household members or MSME employees.                                                     |
| 7    | Zone Classification             | Household: Champion < 3,000 kg, Improver 3,000–6,000 kg, Defaulter > 6,000 kg. MSME: Champion < 6,000 kg, Improver 6,000–12,000 kg. |
| 8    | Recommendations                 | Generate top 3 personalised recommendations (HIGH / MEDIUM / LOW priority) based on the highest-emission categories.                |

## 4.6 Insurance Discount Engine

| Req ID    | Description                                                                                                                                                                     | Priority | Role   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-DIS-01 | Zone discount is applied based on carbon zone: Green Champion earns up to 30%, Improver up to 15%, Defaulter earns 0%.                                                          | High     | System |
| FR-DIS-02 | Improvement bonus is applied if the user reduced their carbon score from the previous year: ≥30% reduction earns +12%, ≥20% earns +8%, ≥10% earns +5%.                          | High     | System |
| FR-DIS-03 | Duration discount is applied for multi-year policies: 2-year policy earns +10%, 1-year earns +5%.                                                                               | Medium   | System |
| FR-DIS-04 | Total combined discount is capped at a maximum of 35%.                                                                                                                          | High     | System |
| FR-DIS-05 | Discount breakdown is shown to the user line by line: Base Premium, Zone Discount, Improvement Bonus, Duration Discount, Final Premium.                                         | High     | User   |
| FR-DIS-06 | Policies page shows base prices to all users. Users with a verified score see personalised discounted prices. Users without a score see a prompt to complete their declaration. | High     | User   |

## 4.7 Insurance Policy Module

| Req ID    | Description                                                                                                                                                                     | Priority | Role   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-POL-01 | Admin manages the insurance policy catalog in the backend. Policies are fetched dynamically — no static policy data on the frontend.                                            | High     | Admin  |
| FR-POL-02 | User browses available policies. If they have a verified carbon score, personalised discount is shown. Otherwise, base prices are shown with a message to complete declaration. | High     | User   |
| FR-POL-03 | User can purchase a policy. Backend validates the purchase, saves a UserPolicy record, and links it to the user account.                                                        | High     | User   |
| FR-POL-04 | User can view all their purchased policies in My Policies section.                                                                                                              | High     | User   |
| FR-POL-05 | System throws PolicyAlreadyPurchasedException if user tries to buy an already active policy of the same type.                                                                   | Medium   | System |

## 4.8 Notification Module

| Req ID    | Description                                                                                                                                                                           | Priority | Role   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-NOT-01 | System creates in-app notifications for all key events: declaration submitted, agent assigned, verification complete, score generated, policy purchased.                              | High     | System |
| FR-NOT-02 | User can mark individual notifications as read by clicking them. Read notifications are removed from the unread list.                                                                 | High     | User   |
| FR-NOT-03 | User can click Mark All as Read to clear all unread notifications at once.                                                                                                            | High     | User   |
| FR-NOT-04 | Read status is persisted in the database. Refreshing the page does not bring back read notifications.                                                                                 | High     | System |
| FR-NOT-05 | System sends email notifications for all key events via SMTP through EmailService. All email templates are HTML-formatted with GreenSure branding.                                    | High     | System |
| FR-NOT-06 | Toast notifications appear in the top-right corner for all user actions. Toasts are colour-coded: green for success, red for error, yellow for warning. Auto-dismiss after 3 seconds. | Medium   | All    |

## 5. Non-Functional Requirements

### 5.1 Security

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- All API endpoints except /auth/login, /auth/register, /auth/forgot-password, /auth/reset-password are protected by JWT authentication
- Passwords are hashed using BCrypt before storage — plain text passwords are never stored
- JWT tokens are signed with a secret key using JJWT and validated on every request
- Role-based access is enforced at the method level using @PreAuthorize — not just at the URL level
- Password reset tokens expire after 15 minutes
- CSRF is disabled as the API is fully stateless

### 5.2 Performance

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- All admin and agent list views must implement pagination — no unbounded data loads
- Angular routes must be lazy-loaded to minimise initial bundle size
- Scheduler tasks must be lightweight and complete within their interval window
- Email sending must be asynchronous — it must not block the main request thread

### 5.3 Usability

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- Form validation must show red borders on invalid fields and auto-scroll to the first error
- Page must scroll to warning messages when validation errors appear
- Toast notifications must provide immediate feedback for every user action
- Loading states must be shown during all API calls with a consistent loading animation
- All error messages must be human-readable — no technical exception stack traces shown to users

### 5.4 Maintainability

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- Frontend components must be placed inside the components folder — not inside feature folders
- All services must be in the core/services folder following consistent naming
- Backend must follow Controller → Service → Repository layered architecture strictly
- All exceptions must be custom named exceptions handled by GlobalExceptionHandler
- No generic Internal Server Error responses — every error must have a specific custom exception

### 5.5 Scalability

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- H2 in-memory database is used for development. The data layer must be PostgreSQL-ready with a single configuration change
- The automatic agent assignment system scales with the number of declarations and agents without any code changes
- Email sending is decoupled from business logic through a single EmailService class

## 6. Email Notification Specification

All emails are sent through a single EmailService using JavaMailSender over SMTP (Gmail, port 587, TLS). All email bodies are HTML-formatted with GreenSure branding.

### 6.1 User Emails

| Trigger               | Subject                                | Key Content                               |
|-----------------------|----------------------------------------|-------------------------------------------|
| Registration          | Welcome to GreenSure                   | Welcome message, login link               |
| Declaration Submitted | Declaration Submitted — GreenSure      | Confirmation, agent assignment within 72h |
| Agent Assigned        | Agent Assigned — GreenSure             | Agent assigned, verification within 72h   |
| Declaration Verified  | Declaration Verified — GreenSure       | Verified, score ready, login to view      |
| Declaration Rejected  | Declaration Rejected — GreenSure       | Rejected, can edit and resubmit           |
| Score Generated       | Your Carbon Score is Ready — GreenSure | Zone, discount %, login to view           |
| Renewal Due           | Time to Renew — GreenSure              | Annual renewal reminder                   |
| Forgot Password       | Reset Your GreenSure Password          | Reset link (expires 15 min)               |
| Password Reset        | Password Reset Successful — GreenSure  | Confirmation, contact if not you          |

### 6.2 Agent Emails

| Trigger                   | Subject                                       | Key Content                             |
|---------------------------|-----------------------------------------------|-----------------------------------------|
| Auto-assigned (Scheduler) | New Declaration Assigned — GreenSure          | Declaration ID, deadline, login link    |
| Manually assigned (Admin) | New Declaration Assigned by Admin — GreenSure | Declaration ID, deadline, login link    |
| Reassigned — new agent    | Declaration Assigned to You — GreenSure       | Declaration ID, new deadline            |
| Reassigned — old agent    | Assignment Removed — GreenSure                | Declaration ID, reason for reassignment |
| Assignment cancelled      | Assignment Cancelled — GreenSure              | Declaration ID, cancelled by admin      |
| 24h deadline warning      | Assignment Deadline Reminder — GreenSure      | Declaration ID, hours remaining         |
| Strike applied            | Strike Applied — GreenSure                    | Declaration ID, current strike count    |

## 7. Custom Exception Specification

All exceptions must be handled by the GlobalExceptionHandler using @RestControllerAdvice. Every error response must use the standard ApiResponse<T> format with a timestamp, HTTP status code, and human-readable message. No generic Internal Server Error responses are permitted.

| Exception Class                       | HTTP Status | When Thrown                                                    |
|---------------------------------------|-------------|----------------------------------------------------------------|
| DeclarationNotFoundException          | 404         | Declaration ID does not exist                                  |
| DeclarationNotEditableException       | 400         | Attempt to edit a non-editable declaration                     |
| DuplicateDeclarationException         | 409         | User tries to submit a second declaration for the same year    |
| MaxResubmissionLimitExceededException | 400         | User exceeds 3 resubmission attempts on a rejected declaration |
| DeclarationAlreadyAssignedException   | 409         | Admin tries to assign an already-assigned declaration          |
| AssignmentNotFoundException           | 404         | Assignment ID does not exist                                   |
| AgentNotFoundException                | 404         | Agent ID does not exist                                        |
| AgentNotAvailableException            | 400         | Selected agent is not available for assignment                 |
| UserNotFoundException                 | 404         | User ID or email does not exist                                |
| PolicyNotFoundException               | 404         | Policy ID does not exist                                       |
| PolicyAlreadyPurchasedException       | 409         | User tries to buy an already active policy                     |
| InvalidTokenException                 | 400         | JWT or password reset token is invalid or expired              |
| AccountSuspendedException             | 403         | Suspended user tries to login                                  |
| CarbonScoreNotFoundException          | 404         | Score not found for user or declaration                        |
| DuplicateScoreException               | 409         | Score already exists for this declaration                      |

## 8. API Endpoint Summary

### 8.1 Auth Endpoints

| Method | Endpoint              | Access | Description                    |
|--------|-----------------------|--------|--------------------------------|
| POST   | /auth/register        | Public | Register new user (HH or MSME) |
| POST   | /auth/login           | Public | Login and receive JWT token    |
| POST   | /auth/forgot-password | Public | Request password reset email   |
| POST   | /auth/reset-password  | Public | Reset password using token     |

### 8.2 Declaration Endpoints

| Method | Endpoint                       | Access | Description                           |
|--------|--------------------------------|--------|---------------------------------------|
| POST   | /declaration/submit            | USER   | Submit new carbon declaration         |
| PUT    | /declaration/{id}/edit         | USER   | Edit draft or rejected declaration    |
| GET    | /declaration/my                | USER   | Get all declarations for current user |
| GET    | /declaration/{id}              | USER   | Get single declaration details        |
| GET    | /declaration/history           | USER   | Declaration history with status       |
| GET    | /admin/declarations            | ADMIN  | All declarations platform-wide        |
| GET    | /admin/declarations/unassigned | ADMIN  | Declarations with no agent assigned   |
| GET    | /admin/declarations/assigned   | ADMIN  | Declarations with active assignment   |

### 8.3 Assignment Endpoints

| Method | Endpoint                       | Access | Description                             |
|--------|--------------------------------|--------|-----------------------------------------|
| GET    | /admin/agents/available        | ADMIN  | List agents available for assignment    |
| POST   | /admin/assignment/assign       | ADMIN  | Manually assign agent to declaration    |
| PUT    | /admin/assignment/reassign     | ADMIN  | Reassign declaration to different agent |
| PUT    | /admin/assignment/change-agent | ADMIN  | Change agent by assignment ID           |
| DELETE | /admin/assignment/cancel/{id}  | ADMIN  | Cancel active assignment                |

## 9. Enhancements & Future Scope

### 9.1 Pending Enhancements (Next Sprint)

| Req ID | Description                                                                                                                                | Priority | Role   |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| ENH-01 | Improve carbon score formula with more accurate CO <sub>2</sub> coefficients and additional input fields aligned to real carbon standards. | High     | System |
| ENH-02 | Implement pagination on all admin and agent list views.                                                                                    | High     | Admin  |
| ENH-03 | Fix frontend file structure — all components inside components folder, all services in core/services.                                      | Medium   | Dev    |
| ENH-04 | Add file upload for supporting documents (electricity bills, vehicle RC, solar proof). Agent can verify uploaded documents.                | High     | Agent  |
| ENH-05 | Implement complete insurance flow — user browses, buys policy, agent acts as underwriter, claims officer handles claims.                   | High     | User   |
| ENH-06 | Research and align carbon score methodology with real-world green insurance and net zero standards.                                        | Medium   | Dev    |

### 9.2 Future Scope — Roadmap

#### Next 3 Months — Production Ready

- Replace H2 with PostgreSQL for production database
- Deploy backend on AWS or Railway
- Deploy Angular frontend on Vercel or Netlify
- Angular HTTP interceptors to remove manual JWT header injection
- Full custom exception hierarchy across all services

#### Next 6 Months — Scale & Compliance

- Mobile app using Angular and Capacitor
- Real IRDAI-compliant insurance policy integration
- Google Maps API for agent proximity matching instead of PIN code
- Carbon score public API for third-party platform integration
- Bulk CSV declaration upload for large MSMEs
- Admin live carbon heatmap with real chart visualisation

#### Long Term Vision — Intelligence & Ecosystem

- ML model for carbon declaration anomaly detection before agent visit
- IoT integration — smart meter auto-populates electricity data in declaration
- Carbon credit marketplace — Champion users earn tradeable carbon credits
- Commercial vehicle fleet insurance expansion
- Open banking integration — utility bills auto-imported into declaration
- National carbon registry integration

## 10. Core Data Model

The following 12 core entities form the GreenSure data model:

| Entity             | Key Fields & Purpose                                                                                                                        |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| User               | id, name, email, password (BCrypt), role, userType, isActive, resubmission_count, passwordResetToken, passwordResetTokenExpiry              |
| HouseholdProfile   | id, user_id, numberOfMembers, address, pinCode, city, state                                                                                 |
| MsmeProfile        | id, user_id, businessName, numberOfEmployees, businessType, address, pinCode                                                                |
| CarbonDeclaration  | id, user_id, declarationYear, status (DRAFT/SUBMITTED/UNDER_VERIFICATION/VERIFIED/REJECTED), resubmissionCount, submittedAt                 |
| DeclarationVehicle | id, declaration_id, vehicleType, fuelType, kmPerYear, vehicleCount                                                                          |
| AgentAssignment    | id, declaration_id, agent_id, assignedBy (SYSTEM/ADMIN), assignmentStatus (ACTIVE/COMPLETED/REASSIGNED/CANCELLED), deadline, reassignReason |
| Verification       | id, assignment_id, agent_id, outcome (CONFIRMED/MODIFIED/REJECTED), rejectionReason, gpsLat, gpsLng, verifiedAt                             |
| CarbonScore        | id, user_id, declaration_id, totalCo2, energyCo2, transportCo2, lifestyleCo2, perCapitaCo2, zone, discountPercent, generatedAt              |
| Policy             | id, name, description, basePrice, policyType, isActive                                                                                      |
| UserPolicy         | id, user_id, policy_id, purchasedAt, discountApplied, finalPrice, status                                                                    |
| Recommendation     | id, user_id, declaration_id, category, priority (HIGH/MEDIUM/LOW), description, generatedAt                                                 |
| Notification       | id, user_id, message, isRead, createdAt, notificationType                                                                                   |